

NDPK is an enzyme in budding yeast that uses ATP to phosphorylate nucleotide diphosphates (NDPs and dNDPs), yielding the triphosphate form of each (NTPs and dNTPs). dNTPs are necessary for DNA synthesis, yet yeast cells that lack NDPK grow and divide at the same rate as wild-type yeast. Therefore, triphosphate generation is most likely carried out by one or more other enzymes.

The enzyme TMPK converts deoxythymidine monophosphate (dTMP) to the diphosphate form (dTDP) by transferring the γ -phosphate of ATP to dTMP. Yeast cells that lack TMPK activity cannot synthesize DNA and become arrested during replication. It was proposed that TMPK may partially compensate for NDPK loss by phosphorylating dTDP as well as dTMP.

To better understand the role of TMPK in nucleotide phosphorylation, wild-type TMPK and the inactive E75K variant were purified by affinity chromatography, and the following experiments were performed.

Experiment 1

After purification, a sample containing 3 μ g of protein was added to a reaction mixture containing radiolabeled nucleotide monophosphates (NMPs or dNMPs) and unlabeled ATP. The reaction proceeded for 30 minutes, and products were separated by thin-layer chromatography (TLC) on a PEI-cellulose plate. PEI-cellulose separates nucleotides by charge (Figure 1), and migration distance decreases with increasing negative charge. Spots were visualized by autoradiography, and it was determined that 0.30 nmol of dTMP and 0.15 nmol of uridine monophosphate (UMP) became phosphorylated, respectively.

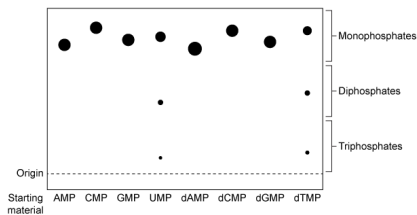


Figure 1 TLC analysis of nucleotide monophosphates after exposure to TMPK and ATP. (Note: Different nucleotides with the same phosphorylation level migrate different distances.)

Experiment 2

The experiment was repeated using radiolabeled dTDP as the starting material. The reaction proceeded in buffer only, in the presence of E75K TMPK, or in the presence of wild-type TMPK. Products were separated by TLC and visualized (Figure 2).

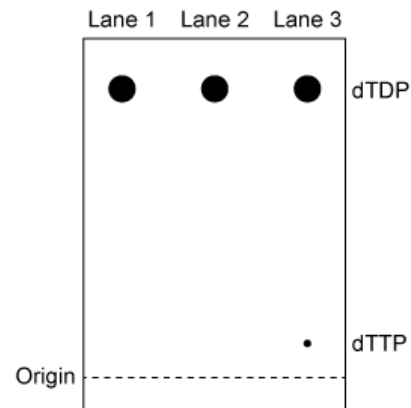


Figure 2 TLC analysis of dTDP phosphorylation with buffer only (Lane 1), E75K TMPK (Lane 2), or wild-type TMPK (Lane 3)

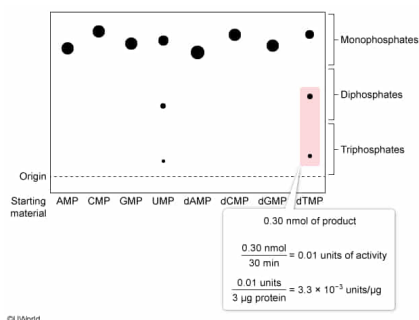
Given that 1 unit of activity equals 1 nmol of substrate converted per minute, what was the specific activity of purified TMPK toward dTMP in Experiment 1? (Note: Assume the rate of phosphorylation is constant throughout the experiment.)

- ☐ A. 1.7×10^{-3} units/ μg [4%]
- ✓ ☒ B. 3.3×10^{-3} units/ μg [50%]
- ☐ C. 5.0×10^{-3} units/ μg [6%]
- ☐ D. 1.0×10^{-2} units/ μg [38%]

Correct

50% answered correctly

Explanation:



The **activity of an enzyme**, given in nanomoles (nmol) of substrate converted to product per minute, is a measure of the **total amount of enzyme** present in a solution. **Specific activity** is the **activity per microgram** of protein in solution. Because the solution may contain proteins other than the enzyme of interest, specific activity is a measure of **enzyme purity**.

According to the passage, 0.30 nmol of dTMP became phosphorylated over the course of the 30-minute assay. Because a unit of activity is defined as 1 nmol/min in this case, the activity of this solution can be calculated as

$$\text{activity} = \frac{0.30 \text{ nmol}}{30 \text{ min}} = \frac{0.01 \text{ nmol}}{\text{min}} = 0.01 \text{ units}$$

To convert the total activity to specific activity, this result must be divided by the total amount of protein in solution. The passage states that Experiment 1 was conducted with 3 μg of purified protein. Therefore, the specific activity of the purified TMPK is

$$\frac{0.01 \text{ units}}{3 \mu\text{g}} = \frac{3.3 \times 10^{-3} \text{ units}}{\mu\text{g}}$$

(Choice A) 1.7×10^{-3} units/ μg is the specific activity of TMPK toward UMP (0.15 nmol of product), not dTMP.

(Choice C) 5.0×10^{-3} units/ μg would be the specific activity of TMPK toward UMP (not dTMP) if only 1 μg of protein were required to achieve the observed reaction rate.

(Choice D) 1.0×10^{-2} units/ μg would be the specific activity of TMPK toward dTMP if only 1 μg of protein were required to achieve the observed reaction rate.

Educational objective:

The activity of an enzyme solution indicates the total amount of enzyme present in a solution by measuring the amount of substrate converted to product per minute. Specific activity is the activity generated by a defined amount of protein (eg, 1 μg). The specific activity of a protein solution indicates the purity of the enzyme of interest.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402447
System:
Enzymes

NDPK is an enzyme in budding yeast that uses ATP to phosphorylate nucleotide diphosphates (NDPs and dNDPs), yielding the triphosphate form of each (NTPs and dNTPs). dNTPs are necessary for DNA synthesis, yet yeast cells that lack NDPK grow and divide at the same rate as wild-type yeast. Therefore, triphosphate generation is most likely carried out by one or more other enzymes.

The enzyme TMPK converts deoxythymidine monophosphate (dTMP) to the diphosphate form (dTDP) by transferring the γ -phosphate of ATP to dTMP. Yeast cells that lack TMPK activity cannot synthesize DNA and become arrested during replication. It was proposed that TMPK may partially compensate for NDPK loss by phosphorylating dTDP as well as dTMP.

To better understand the role of TMPK in nucleotide phosphorylation, wild-type TMPK and the inactive E75K variant were purified by affinity chromatography, and the following experiments were performed.

Experiment 1

After purification, a sample containing 3 μ g of protein was added to a reaction mixture containing radiolabeled nucleotide monophosphates (NMPs or dNMPs) and unlabeled ATP. The reaction proceeded for 30 minutes, and products were separated by thin-layer chromatography (TLC) on a PEI-cellulose plate. PEI-cellulose separates nucleotides by charge (Figure 1), and migration distance decreases with increasing negative charge. Spots were visualized by autoradiography, and it was determined that 0.30 nmol of dTMP and 0.15 nmol of uridine monophosphate (UMP) became phosphorylated, respectively.

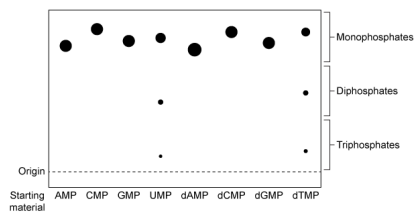


Figure 1 TLC analysis of nucleotide monophosphates after exposure to TMPK and ATP. (Note: Different nucleotides with the same phosphorylation level migrate different distances.)

Experiment 2

The experiment was repeated using radiolabeled dTDP as the starting material. The reaction proceeded in buffer only, in the presence of E75K TMPK, or in the presence of wild-type TMPK. Products were separated by TLC and visualized (Figure 2).

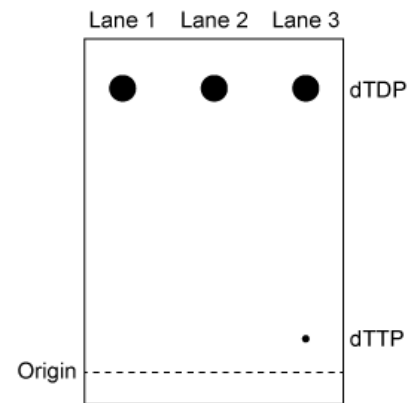


Figure 2 TLC analysis of dTDP phosphorylation with buffer only (Lane 1), E75K TMPK (Lane 2), or wild-type TMPK (Lane 3)

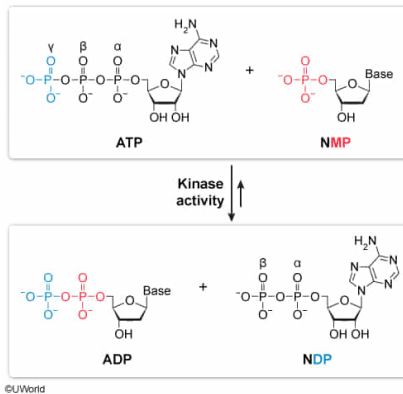
NDPK and TMPK are members of a class of enzymes called:

- ☐ A. phosphatases. [4%]
- ✓ ☒ B. kinases. [83%]
- ☐ C. phosphorylases. [11%]
- ☐ D. ligases.

Correct

82% answered correctly

Explanation:



Several different classes of enzyme **add phosphates** onto molecules or **remove phosphates** from them. These enzymes are classified based on the mechanisms they use to transfer phosphates.

Kinases transfer the γ -phosphate from NTPs (usually ATP or GTP) to other molecules to yield an NDP and phosphorylated substrate. Kinases can also catalyze the reverse reaction, in which a phosphate is transferred from a biomolecule to an NDP to form the dephosphorylated molecule and an NTP. For example, phosphoglycerate kinase transfers a phosphate from ATP to 3-phosphoglycerate to produce 1,3-bisphosphoglycerate during [gluconeogenesis](#), but it catalyzes the reverse reaction during [glycolysis](#).

The passage states that NDPK uses ATP to phosphorylate NDPs and that TMPK transfers the γ -phosphate of ATP to dTMP. Therefore, **NDPK and TMPK are kinases**.

(Choice A) [Phosphatases](#) remove phosphate groups from molecules by hydrolysis, yielding the dephosphorylated substrate and inorganic phosphate (P_i). NDPK and TMPK neither produce nor consume P_i , so they are not phosphatases.

(Choice C) [Phosphorylases](#) break bonds by adding P_i across them. NDPK and TMPK use ATP rather than P_i , so they are not phosphorylases.

(Choice D) [Ligases](#) link two molecules using NTP hydrolysis as an energy source, releasing P_i . NDPK and TMPK transfer a phosphate from one molecule to another and do not release P_i .

Educational objective:

Several classes of enzymes alter the phosphorylation state of molecules. Kinases transfer phosphate groups from NTPs to other molecules (or from other molecules to NDPs), phosphatases remove phosphate groups by hydrolysis, phosphorylases break bonds by adding inorganic phosphate across them, and ligases hydrolyze NTPs for the energy necessary to link two molecules together.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402448
System:
Enzymes

NDPK is an enzyme in budding yeast that uses ATP to phosphorylate nucleotide diphosphates (NDPs and dNDPs), yielding the triphosphate form of each (NTPs and dNTPs). dNTPs are necessary for DNA synthesis, yet yeast cells that lack NDPK grow and divide at the same rate as wild-type yeast. Therefore, triphosphate generation is most likely carried out by one or more other enzymes.

The enzyme TMPK converts deoxythymidine monophosphate (dTMP) to the diphosphate form (dTDP) by transferring the γ -phosphate of ATP to dTMP. Yeast cells that lack TMPK activity cannot synthesize DNA and become arrested during replication. It was proposed that TMPK may partially compensate for NDPK loss by phosphorylating dTDP as well as dTMP.

To better understand the role of TMPK in nucleotide phosphorylation, wild-type TMPK and the inactive E75K variant were purified by affinity chromatography, and the following experiments were performed.

Experiment 1

After purification, a sample containing 3 μ g of protein was added to a reaction mixture containing radiolabeled nucleotide monophosphates (NMPs or dNMPs) and unlabeled ATP. The reaction proceeded for 30 minutes, and products were separated by thin-layer chromatography (TLC) on a PEI-cellulose plate. PEI-cellulose separates nucleotides by charge (Figure 1), and migration distance decreases with increasing negative charge. Spots were visualized by autoradiography, and it was determined that 0.30 nmol of dTMP and 0.15 nmol of uridine monophosphate (UMP) became phosphorylated, respectively.

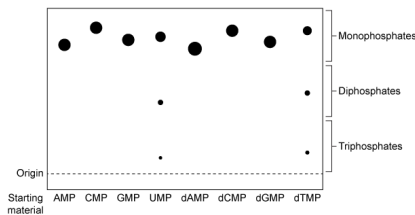


Figure 1 TLC analysis of nucleotide monophosphates after exposure to TMPK and ATP. (Note: Different nucleotides with the same phosphorylation level migrate different distances.)

Experiment 2

The experiment was repeated using radiolabeled dTDP as the starting material. The reaction proceeded in buffer only, in the presence of E75K TMPK, or in the presence of wild-type TMPK. Products were separated by TLC and visualized (Figure 2).

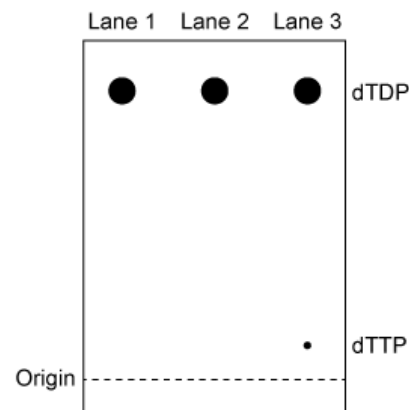


Figure 2 TLC analysis of dTDP phosphorylation with buffer only (Lane 1), E75K TMPK (Lane 2), or wild-type TMPK (Lane 3)

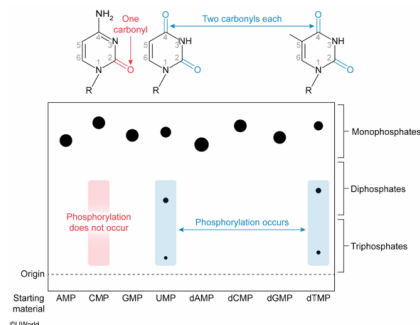
Which statement about TMPK specificity is supported by Figure 1?

- ☐ A. TMPK can modify the monophosphate form of nucleotides but not the diphosphate form. [14%]
- ☐ B. TMPK can interact only with nucleotides that have methyl groups. [13%]
- ✓ ☒ C. TMPK can phosphorylate only pyrimidine nucleotides that have two carbonyls. [67%]
- ☐ D. TMPK can phosphorylate dNMPs but not NMPs. [4%]

Correct

67% answered correctly

Explanation:



Different enzymes have **different levels of specificity** toward their substrates. Some enzymes are **highly specific** and will catalyze reactions only with a **single molecule** or with a few **highly similar molecules**. Other enzymes are **less specific** and will catalyze reactions with **several different molecules** that have related structures. The specificity of an enzyme can be **qualitatively determined** by observing its ability to act on related molecules.

Figure 1 shows that TMPK can phosphorylate the pyrimidine nucleotides dTMP and UMP, but it does not phosphorylate any of the other nucleotides tested. Cytidine monophosphate (CMP) is a pyrimidine nucleotide that is identical to UMP except that CMP has an amino group instead of a carbonyl at position 4 and does not have a hydrogen atom bonded to the nitrogen at position 3.

The fact that TMPK can phosphorylate UMP but not the nearly identical CMP indicates that the enzyme is specific to the aspect of UMP that differs from CMP. Both dTMP and UMP have two carbonyls whereas CMP has only one.

Consequently, the statement that **TMPK can phosphorylate only pyrimidine nucleotides that have two carbonyls** is supported by the data in Figure 1.

(Choice A) Figure 1 shows that TMPK can phosphorylate both dTMP (forming the diphosphate spot) and dTDP (forming the triphosphate spot). Therefore, TMPK can modify both the monophosphate *and* diphosphate forms of nucleotides.

(Choice B) Figure 1 shows that TMPK can phosphorylate UMP, which, unlike dTMP, does not contain a methyl group. Therefore, TMPK can interact with at least one molecule that does not contain a methyl group.

(Choice D) Figure 1 shows that TMPK can phosphorylate both dTMP and UMP. Therefore, TMPK can phosphorylate both the *deoxy-* form (dNMPs) and the *oxy-* form (NMPs) of nucleotides.

Educational objective:

Different enzymes have different levels of specificity for their substrates. Enzymes with high specificity will modify only one molecule or a few highly similar molecules, whereas less specific enzymes modify several molecules within a particular class. An enzyme's specificity can be determined qualitatively by observing whether it acts on similar substrates.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402449
System:
Enzymes

NDPK is an enzyme in budding yeast that uses ATP to phosphorylate nucleotide diphosphates (NDPs and dNDPs), yielding the triphosphate form of each (NTPs and dNTPs). dNTPs are necessary for DNA synthesis, yet yeast cells that lack NDPK grow and divide at the same rate as wild-type yeast. Therefore, triphosphate generation is most likely carried out by one or more other enzymes.

The enzyme TMPK converts deoxythymidine monophosphate (dTMP) to the diphosphate form (dTDP) by transferring the γ -phosphate of ATP to dTMP. Yeast cells that lack TMPK activity cannot synthesize DNA and become arrested during replication. It was proposed that TMPK may partially compensate for NDPK loss by phosphorylating dTDP as well as dTMP.

To better understand the role of TMPK in nucleotide phosphorylation, wild-type TMPK and the inactive E75K variant were purified by affinity chromatography, and the following experiments were performed.

Experiment 1

After purification, a sample containing 3 μ g of protein was added to a reaction mixture containing radiolabeled nucleotide monophosphates (NMPs or dNMPs) and unlabeled ATP. The reaction proceeded for 30 minutes, and products were separated by thin-layer chromatography (TLC) on a PEI-cellulose plate. PEI-cellulose separates nucleotides by charge (Figure 1), and migration distance decreases with increasing negative charge. Spots were visualized by autoradiography, and it was determined that 0.30 nmol of dTMP and 0.15 nmol of uridine monophosphate (UMP) became phosphorylated, respectively.

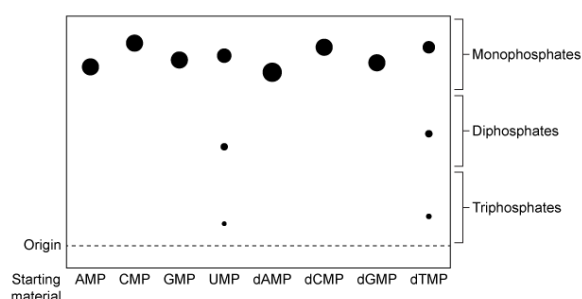


Figure 1 TLC analysis of nucleotide monophosphates after exposure to TMPK and ATP. (Note: Different nucleotides with the same phosphorylation level migrate different distances.)

Experiment 2

The experiment was repeated using radiolabeled dTDP as the starting material. The reaction proceeded in buffer only, in the presence of E75K TMPK, or in the presence of wild-type TMPK. Products were separated by TLC and visualized (Figure 2).

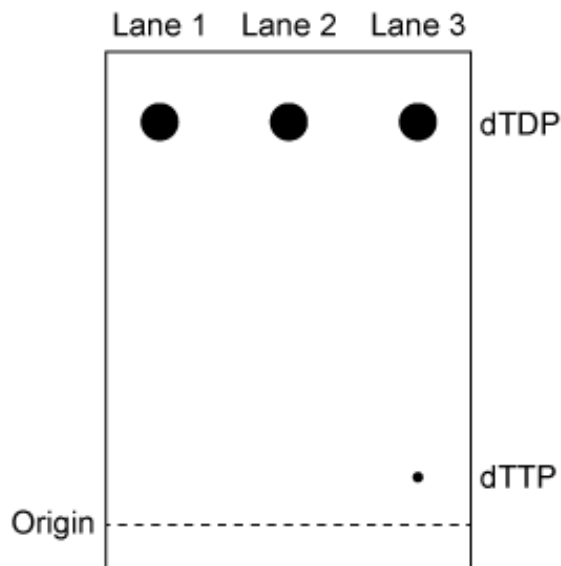


Figure 2 TLC analysis of dTDP phosphorylation with buffer only (Lane 1), E75K TMPK (Lane 2), or wild-type TMPK (Lane 3)

A Mg^{2+} ion in the active site of TMPK is directly involved in catalysis. Given this, Mg^{2+} most likely plays which of the following roles in TMPK-mediated reactions?

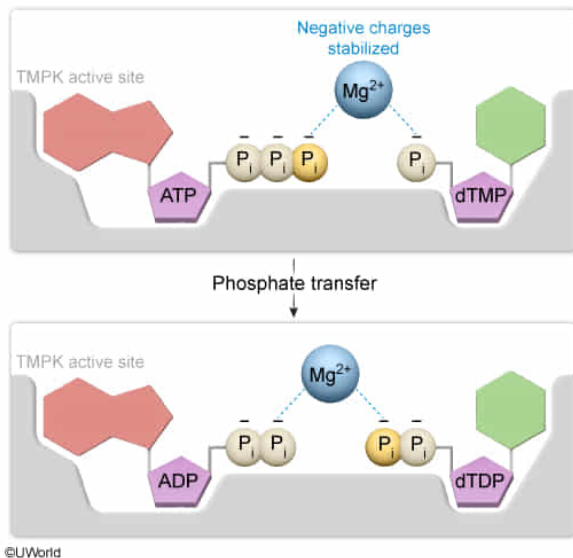
- ☐ A. It binds negatively charged amino acids, stabilizing the tertiary structure of TMPK. [6%]
- ☐ B. It interacts with water molecules in the active site, facilitating phosphate hydrolysis. [5%]
- ☐ C. It binds anions, preventing them from competing with phosphate groups for the active site. [11%]
- ✓ ☒ D. It interacts with the negative charges of phosphate groups, stabilizing the transition state. [75%]

Correct

75% answered correctly

Explanation:

Magnesium in TMPK catalysis



Catalysis is the **acceleration of a reaction** that occurs when a catalyst **stabilizes the transition state** between the reactant and the product. The stabilization of the transition state **decreases the activation energy**, or the energy necessary to initiate the reaction. **Enzymes** are **biological catalysts** that are primarily composed of amino acids but may require other components, known as **cofactors**, for catalysis.

The question states that Mg^{2+} is directly involved in TMPK-mediated catalysis, so Mg^{2+} must be involved in stabilization of the transition state. Phosphate group transfer requires that ATP and dTMP form a high-energy–transition state in which their negative charges repel each other. The positively charged Mg^{2+} ion stabilizes this transition state by interacting with the negatively charged phosphates. As a result, the activation energy for the reaction is decreased in the presence of TMPK, and the reaction rate increases.

(Choice A) The question states that Mg^{2+} is directly involved in catalysis. An enzyme may have a stable tertiary structure (conformational stability) but still be catalytically inactive if it lacks the necessary cofactors.

(Choice B) **Hydrolysis** is the cleavage of a bond by adding water across it. TMPK catalyzes a phosphate transfer reaction, not a phosphate hydrolysis reaction.

(Choice C) Mg^{2+} is directly involved in catalysis, so it must interact with the substrate directly. Preventing anions from competing for the active site could prevent inhibition but would not directly facilitate catalysis.

Educational objective:

Catalysis is the increase in a reaction rate that occurs when a catalyst decreases the activation energy of a reaction. Catalysts decrease activation energy by interacting with substrates to stabilize the transition state between reactant and product. Enzymes are biological catalysts, and some enzymes require cofactors such as metal ions for

catalysis.

Time spent:0 sec

Subject:
Biochemistry

QID:402450

System:
Enzymes

NDPK is an enzyme in budding yeast that uses ATP to phosphorylate nucleotide diphosphates (NDPs and dNDPs), yielding the triphosphate form of each (NTPs and dNTPs). dNTPs are necessary for DNA synthesis, yet yeast cells that lack NDPK grow and divide at the same rate as wild-type yeast. Therefore, triphosphate generation is most likely carried out by one or more other enzymes.

The enzyme TMPK converts deoxythymidine monophosphate (dTMP) to the diphosphate form (dTDP) by transferring the γ -phosphate of ATP to dTMP. Yeast cells that lack TMPK activity cannot synthesize DNA and become arrested during replication. It was proposed that TMPK may partially compensate for NDPK loss by phosphorylating dTDP as well as dTMP.

To better understand the role of TMPK in nucleotide phosphorylation, wild-type TMPK and the inactive E75K variant were purified by affinity chromatography, and the following experiments were performed.

Experiment 1

After purification, a sample containing 3 μ g of protein was added to a reaction mixture containing radiolabeled nucleotide monophosphates (NMPs or dNMPs) and unlabeled ATP. The reaction proceeded for 30 minutes, and products were separated by thin-layer chromatography (TLC) on a PEI-cellulose plate. PEI-cellulose separates nucleotides by charge (Figure 1), and migration distance decreases with increasing negative charge. Spots were visualized by autoradiography, and it was determined that 0.30 nmol of dTMP and 0.15 nmol of uridine monophosphate (UMP) became phosphorylated, respectively.

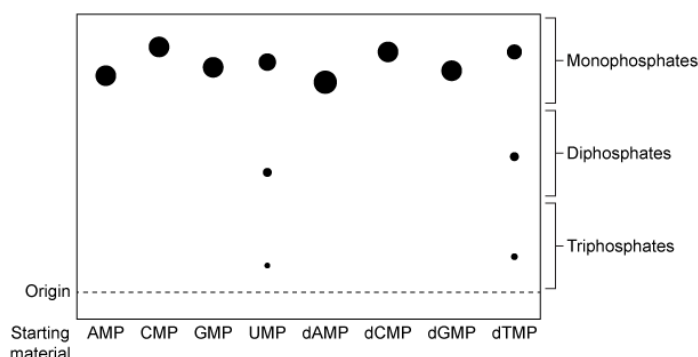


Figure 1 TLC analysis of nucleotide monophosphates after exposure to TMPK and ATP. (Note: Different nucleotides with the same phosphorylation level migrate different distances.)

Experiment 2

The experiment was repeated using radiolabeled dTDP as the starting material. The reaction proceeded in buffer only, in the presence of E75K

TMPK, or in the presence of wild-type TMPK. Products were separated by TLC and visualized (Figure 2).

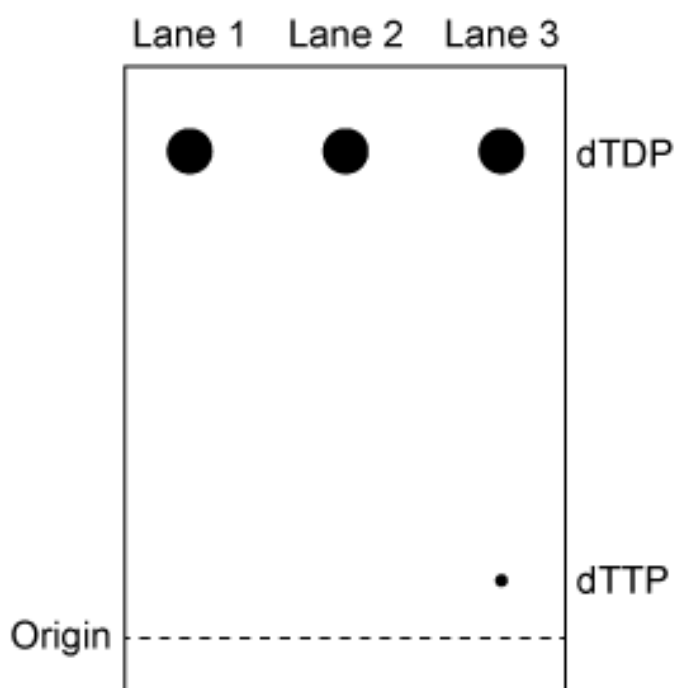


Figure 2 TLC analysis of dTDP phosphorylation with buffer only (Lane 1), E75K TMPK (Lane 2), or wild-type TMPK (Lane 3)

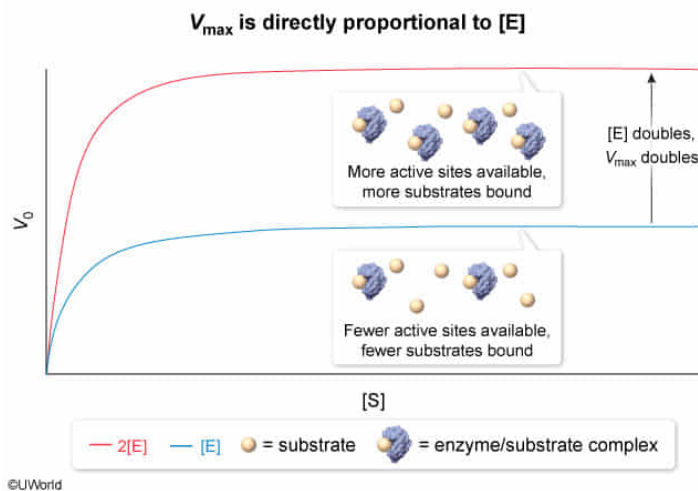
Assuming the reaction in Experiment 2 is performed under saturating conditions, which of the following would cause an increase in the rate of phosphorylation?

- ✓ ☒ A. Increasing the concentration of TMPK [69%]
- ☐ B. Introducing a mutation to TMPK that increases its K_M [11%]
- ☐ C. Allowing the reaction to run for a longer period of time [4%]
- ☐ D. Adding more dTDP to the reaction [14%]

Correct

68% answered correctly

Explanation:



The rate of an enzymatic reaction is described mathematically by the Michaelis-Menten equation:

$$V_0 = \frac{V_{\max}[S]}{K_M + [S]}$$

where V_0 is the observed reaction rate, $V_{\max}$ is the maximum possible reaction rate, $[S]$ is the concentration of substrate present, and K_M is the Michaelis constant. $V_{\max}$ is achieved when $[S]$ is high enough that all enzyme active sites are bound by substrate, a condition known as **saturation**. $V_{\max}$ is **directly proportional** to the **concentration of enzyme** in solution, expressed mathematically as

$$V_{\max} = k_{\text{cat}}[E]$$

where $[E]$ is the total enzyme concentration and k_{cat} is the turnover number, or the number of reactions catalyzed per second per enzyme at saturation. At saturation, **increasing $[S]$ will not increase the reaction rate** because $V_{\max}$ has already been achieved. **Increasing the enzyme concentration**, on the other hand, increases the number of active sites present in solution, and therefore **increases $V_{\max}$** . If Experiment 2 was performed under saturating conditions, increasing the concentration of

the enzyme TMPK would increase the total number of TMPK enzymes bound by substrate at any given time, which in turn would increase the rate of the reaction.

(Choice B) K_M is the concentration of substrate at which $\frac{1}{2}V_{\max}$ is achieved. An **increase in K_M** would increase the amount of substrate needed to achieve $V_{\max}$ but most likely would not alter $V_{\max}$ itself.

(Choice C) Reaction rate is calculated as the amount of product formed divided by the amount of time required for product to form. A longer reaction time would result in more total product but would not increase the *rate* of the reaction.

(Choice D) Under saturating conditions, adding more substrate to the reaction will not increase the reaction rate because all active sites are already occupied. dTDP is the substrate in Experiment 2, so adding more dTDP will not alter the reaction rate.

Educational objective:

The rate of an enzymatic reaction is directly proportional to the concentration of enzyme present in the reaction. When the active sites on an enzyme are saturated, an increase in substrate concentration will not change the reaction rate, but an increase in enzyme concentration will.

Time spent:0 sec

QID:402451

Subject:
Biochemistry

System:
Enzymes

NDPK is an enzyme in budding yeast that uses ATP to phosphorylate nucleotide diphosphates (NDPs and dNDPs), yielding the triphosphate form of each (NTPs and dNTPs). dNTPs are necessary for DNA synthesis, yet yeast cells that lack NDPK grow and divide at the same rate as wild-type yeast. Therefore, triphosphate generation is most likely carried out by one or more other enzymes.

The enzyme TMPK converts deoxythymidine monophosphate (dTMP) to the diphosphate form (dTDP) by transferring the γ -phosphate of ATP to dTMP. Yeast cells that lack TMPK activity cannot synthesize DNA and become arrested during replication. It was proposed that TMPK may partially compensate for NDPK loss by phosphorylating dTDP as well as dTMP.

To better understand the role of TMPK in nucleotide phosphorylation, wild-type TMPK and the inactive E75K variant were purified by affinity chromatography, and the following experiments were performed.

Experiment 1

After purification, a sample containing 3 μg of protein was added to a reaction mixture containing radiolabeled nucleotide monophosphates (NMPs or dNMPs) and unlabeled ATP. The reaction proceeded for 30 minutes, and products were separated by thin-layer chromatography (TLC) on a PEI-cellulose plate. PEI-cellulose separates nucleotides by charge (Figure 1), and migration distance decreases with increasing negative charge. Spots were visualized by autoradiography, and it was determined that 0.30 nmol of dTMP and 0.15 nmol of uridine monophosphate (UMP) became phosphorylated, respectively.

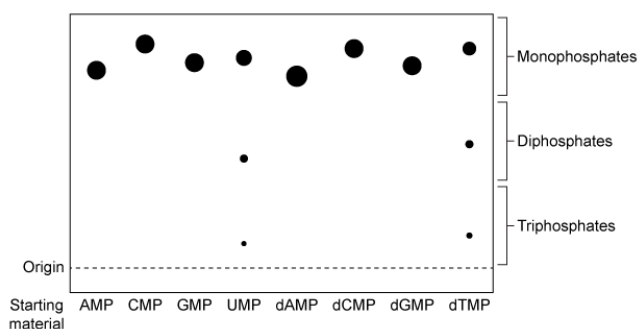


Figure 1 TLC analysis of nucleotide monophosphates after exposure to TMPK and ATP. (Note: Different nucleotides with the same phosphorylation level migrate different distances.)

Experiment 2

The experiment was repeated using radiolabeled dTDP as the starting material. The reaction proceeded in buffer only, in the presence of E75K TMPK, or in the presence of wild-type TMPK. Products were separated by TLC and visualized (Figure 2).

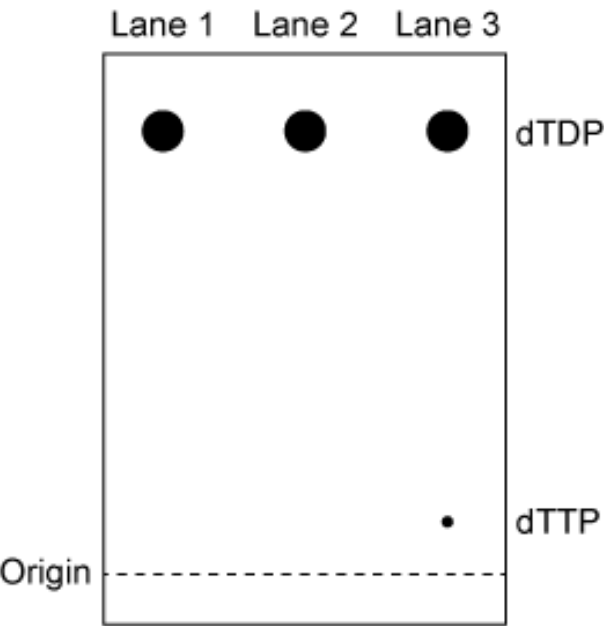
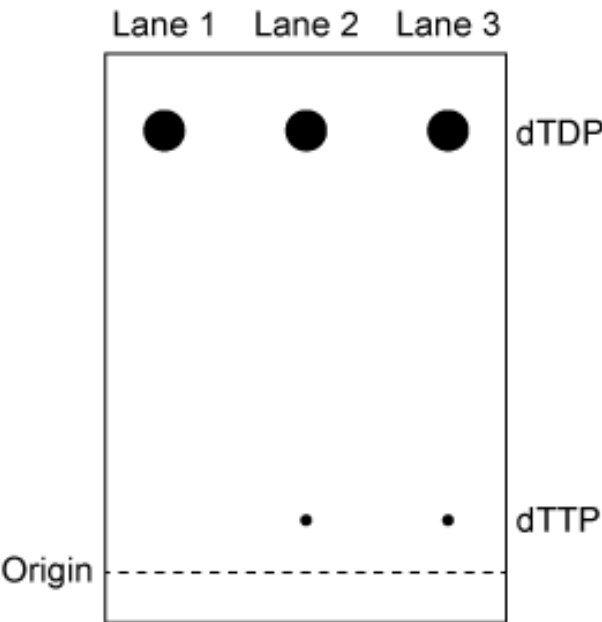


Figure 2 TLC analysis of dTDP phosphorylation with buffer only (Lane 1), E75K TMPK (Lane 2), or wild-type TMPK (Lane 3)

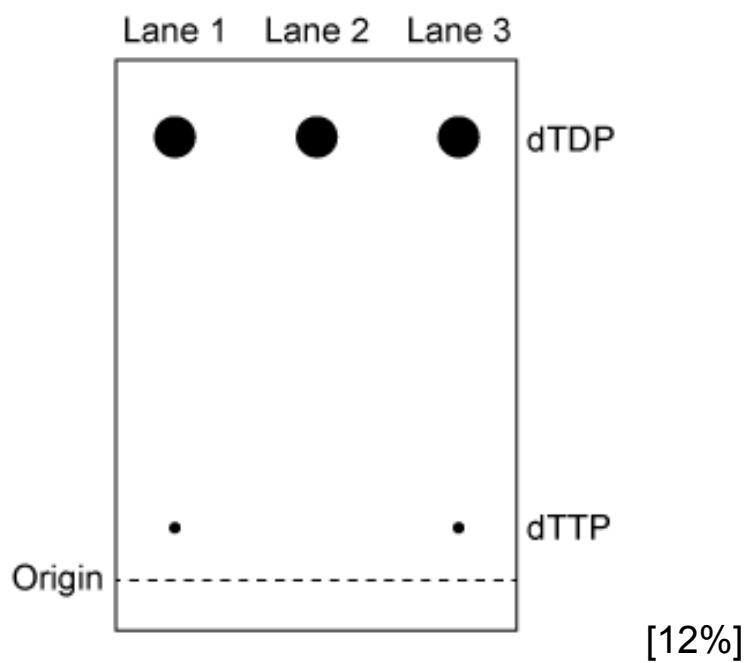
Experiment 2 was performed to determine whether phosphorylation of dTDP was due to residual NDPK activity in the purified enzyme solution. Which of the following shows the expected outcome of Experiment 2 if NDPK were still present after purification?

✔ ☒ A.

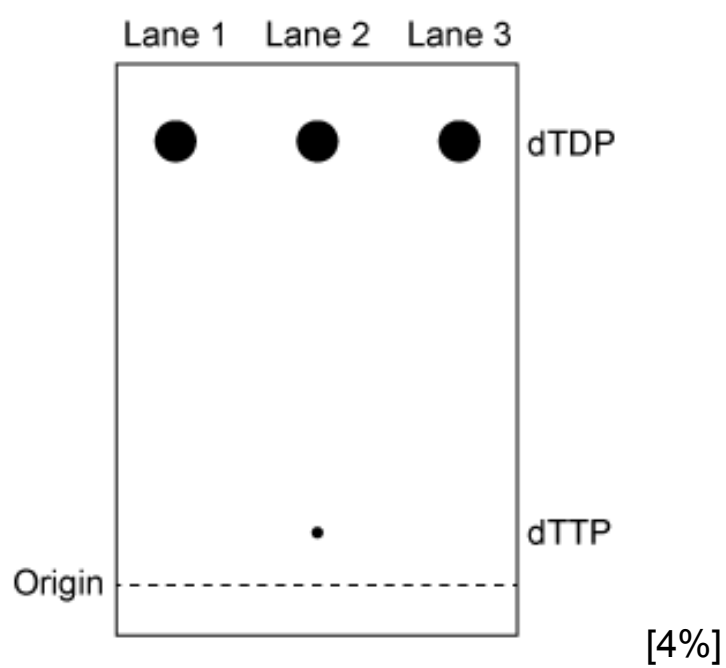


[50%]

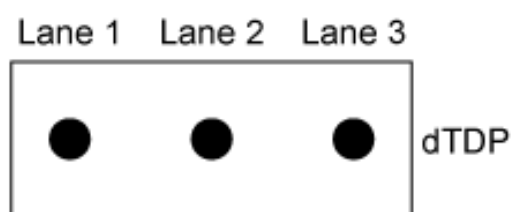
☐ B.

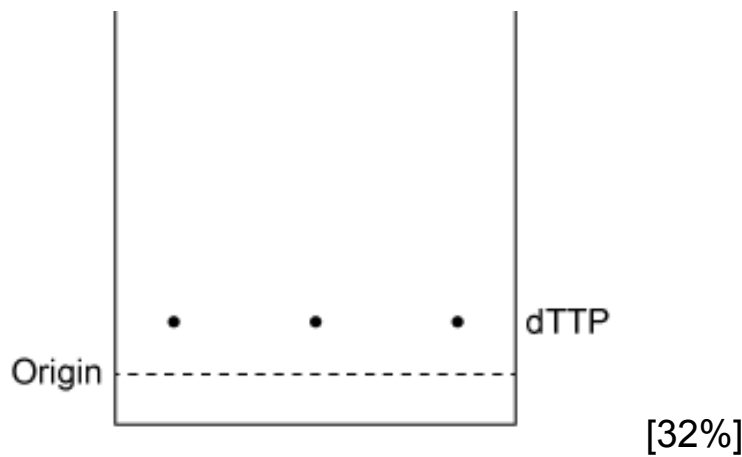


☐ C.



☐ D.

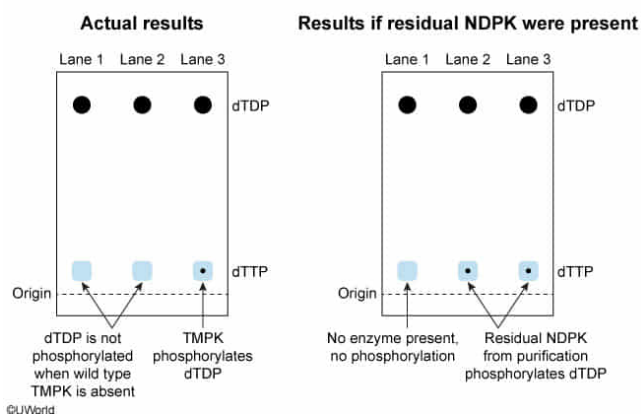




Correct

50% answered correctly

Explanation:



Protein purification separates a protein of interest from all other proteins in a cell culture, allowing the activity of that protein to be assayed. However, **purification techniques are imperfect**, and **trace amounts of proteins** other than the protein of interest may remain in purified solutions.

To confirm that an observed activity is due to the protein of interest, **an inactive form of the protein** can be expressed, purified by the same method as the active form, and assessed for activity. Identical protocols are used to obtain both the wild-type and inactive protein solutions, so they are expected to have the same impurities. Therefore, contributions to the observed activity from impurities such as residual NDPK will be apparent in both solutions.

If phosphorylation of dTDP were due to residual NDPK activity, that activity would also be expected in the purified E75K solution. Therefore, if residual NDPK is the cause of dTTP production, both Lane 2 (E75K) and Lane 3 (wild-type) should have two spots: one at the top that corresponds to unmodified dTDP, and one near the origin that corresponds to newly formed dTTP. The fact that the dTTP spot did *not* appear in Lane 2 of the *actual* experiment (Figure 2)

indicates that there was no residual NDPK, so the purification technique successfully removed NDPK.

(Choices B and D) The reaction in Lane 1 was carried out in buffer only as a negative control (ie, no purified protein was added to the buffer), so no enzymes of any type were present and dTTP will not be observed in this lane. A dTTP spot in Lane 1 would indicate external contamination, and the experiment would have to be repeated.

(Choice C) Lanes 2 and 3 contain identical reaction mixtures, with the exception that the E75K variant of TMPK in Lane 2 is inactive. Therefore, if there is residual NDPK activity, it should be present in both lanes, not just in Lane 2.

Educational objective:

Protein purification is imperfect, and trace amounts of proteins other than the protein of interest may be present in a purified solution. Any observed activity can be attributed to the protein of interest (rather than impurities) if a purified solution of an inactive variant does not yield the same activity.

Time spent:0 sec

Subject:
Biochemistry

QID:402452

System:
Enzymes